

Online Complaint Registration and Management System

Phase 1: Ideation Phase

SmartBridge Internship Project Submission

Team: Online Complaint Registration and Management System

1.1 Brainstorming & Idea Listing

The team conducted a brainstorming session to identify problems faced by citizens, employees, and organizations when reporting and resolving complaints through traditional (manual, phone-based, or in-person) channels. The following ideas were discussed and evaluated:

Idea	Description	Feasibility
Manual Complaint Register	Paper-based complaint books maintained at offices	Low — no tracking, easily lost, slow
Complaint via Phone/Email	Citizens call or email a support desk	Medium — no status visibility, inconsistent records
Mobile App Only Solution	Native Android/iOS app for complaints	Medium — higher dev cost, limited reach for all users
Web-Based Complaint Management System	Full-stack web platform for registration, tracking, and resolution	High — accessible on any device, centralized data, real-time status

Selected Idea: A web-based Online Complaint Registration and Management System built on the MERN stack, allowing users to register complaints, track their status in real time, and interact with assigned agents, while admins manage the full resolution workflow.

1.2 Empathy Map Canvas

An empathy map was created for the primary user persona — a citizen/customer who needs to report an issue — to better understand their needs and pain points.

Category	Observations
Says	"I don't know who to contact." / "I submitted a complaint but have no idea what's happening with it." / "Why do I have to call again and again for updates?"
Thinks	Worried the complaint will be forgotten or lost. Unsure if the right department received it. Doubts whether raising a complaint is even worth the effort.
Does	Visits the office in person or calls repeatedly. Writes complaints on paper or sends emails that go unacknowledged. Asks around informally for updates.
Feels	Frustrated by the lack of transparency. Anxious about delays. Loses trust in the organization when there is no follow-up.

1.3 Literature Survey

Existing complaint-handling approaches and systems were reviewed to identify gaps that this project aims to address.

System / Approach	Strengths	Limitations
Manual complaint registers	Simple, no technical setup required	No tracking, prone to loss, no analytics, slow resolution
Call-center based complaint desks	Human interaction, immediate acknowledgment	No persistent record for the user, inconsistent updates, high operational cost

System / Approach	Strengths	Limitations
Generic ticketing tools (e.g. spreadsheet-based tracking)	Centralized records	No role-based access, no automated notifications, not citizen-facing
Existing MERN-based complaint portals (academic/open-source projects)	Real-time status tracking, role-based dashboards, REST APIs	Often lack agent workflow, analytics dashboards, or production-grade security (this project addresses these gaps)

1.4 Problem Statement

Using the standard problem-statement template:

I am a citizen/customer,

- Trying to report an issue (infrastructure, utilities, sanitation, safety, etc.) and get it resolved,
- But I have no reliable way to track the status of my complaint after submitting it,
- Because complaints are handled manually or through disconnected channels (phone, email, in-person) with no centralized tracking system,
- Which makes me feel frustrated, ignored, and unsure whether my issue will ever be addressed.

Problem Statement (concise): There is no centralized, transparent, and trackable platform for citizens to register complaints, monitor their resolution status, and communicate with the responsible agents — resulting in delayed resolutions, lost records, and reduced accountability for the handling organization.